



CLUSTER INNOVATION CENTRE · CIC-TBI · PROGRESS REVIEW

Quarterly Progress Report

July to September 2026, month by month

TO	Mr Pratik Kumar, Project Manager, CIC-TBI, Cluster Innovation Centre, University of Delhi
FROM	Protopus (Soham, Founder)
SUBMITTED	5 October 2026
FOR THE MEETING	12 October 2026, 2:00 PM, Director's Office, 3rd Floor, Cluster Innovation Centre, University Stadium, Gate No. 2
PERIOD COVERED	1 July to 30 September 2026
LINKS	protopus-kappa.vercel.app (website) · probe-matrix.vercel.app (Probe Matrix concept)

Dates are approximate (early, mid or late in the month). Prices and costs are projections from our cost report and are not yet validated by pilots.

SUMMARY OF THE QUARTER

From a researched idea to a prototype under build, with funding conversations open.

Protopus is building smart machines and accessories for hardware prototyping, guided by an AI engineer called Proto. [Baker](#), the first machine, makes a finished circuit board on the desk. [Mixer](#), the second, tests it. Over the quarter we completed our research and website (July), finalised the prototype design and started building it (August), and integrated software with hardware while opening funding talks for the proof of concept (September). We also conceptualised a new accessory, the [Probe Matrix](#).

Jul	Aug	Sep
research, website, investor meeting	design finalised, build started	integration, funding talks, Probe Matrix

STATUS AT A GLANCE · 30 SEPTEMBER 2026

MILESTONE	MONTH	STATUS
Market and competition research	Jul	DONE
Website live (Baker and Mixer pages)	Jul	DONE
Meeting with Samuel Beek (Schematik): funding offered after the proof of concept	Jul	DONE
Prototype design finalised	Aug	DONE
Build of the first Baker prototype	Aug–	IN PROGRESS
Software–hardware integration	Sep–	IN PROGRESS
Proof-of-concept funding from STMicroelectronics	Sep–	IN DISCUSSION
Probe Matrix accessory	Sep	CONCEPT
Costing, workflow, China manufacturing plan and pitch documents	Sep	DONE



Approximate positions within each month.

Research, website and our first investor conversation.

July was about understanding the problem and the market, and putting Protopus in front of people.

1.1 Market and competition research

EARLY JULY · DONE

We studied how hardware prototypes are made today and where people get stuck: fab orders, 7–20 days for a JLCPCB board to reach India, hand assembly, and test benches that need separate instruments. We also studied the tools that exist (desktop mills, ink printers, pick-and-place machines, USB instruments and AI design tools) and found that each solves only one step.

1.2 Product concept

EARLY TO MID JULY · DONE

We defined the system: Proto, an AI engineer that plans the build; [Baker](#), a machine that makes the board (mill, solder mask, stencil, paste, reflow); and [Mixer](#), an instrument that tests it. We decided the machines are for prototyping, the first 1–10 boards of a design, while volume stays with fabs.

1.3 Website designed and launched

MID JULY · DONE

We published the Protopus website, with pages for [Baker](#) and [Mixer](#): protopus-kappa.vercel.app.

1.4 Meeting with Samuel Beek, Founder and CEO of Schematik

LATE JULY · DONE

We presented the idea to Samuel Beek. He committed to fund Protopus once the proof of concept is achieved. This is a verbal commitment, and it depends on the proof of concept.

Prototype design finalised; the real prototype started.

In August we moved from concept to engineering design and began building.

2.1 Prototype design finalised

EARLY TO MID AUGUST · DONE

We fixed the architecture of the first [Baker](#) prototype. It runs 15 process steps on one shuttle, in two process zones: Zone A mills, drills and laser-cuts the stencil; Zone B coats, images and cures the solder mask. The reflow is low-temperature. Control is a Raspberry Pi 5 with a BTT Octopus Pro board running Klipper. We also defined the [Mixer](#) instrument architecture for a later stage.

2.2 Parts selection for the first prototype

MID AUGUST · DONE

We chose the parts for each subsystem (motion, spindle and laser, mask and UV, reflow, control electronics) so that procurement could start.

2.3 Build of the real prototype started

LATE AUGUST · IN PROGRESS

We started procurement and assembly of the first [Baker](#) prototype.

Integration, proof-of-concept funding, and a new accessory.

In September we worked on making the software and the machine work together, opened funding talks for the proof of concept, and documented the business.

3.1 Software–hardware integration

THROUGH SEPTEMBER · IN PROGRESS

We worked on connecting the software layer to the prototype’s control electronics. That layer is [Baker](#) Studio, where Proto checks a design and plans the machine job, and it sits on the Klipper firmware.

3.2 Raising proof-of-concept funding from STMicroelectronics

MID TO LATE SEPTEMBER · IN DISCUSSION

We approached STMicroelectronics to fund the prototype and the proof of concept. Discussions are ongoing; nothing has been committed yet.

3.3 [Probe Matrix](#) conceptualised

LATE SEPTEMBER, AFTER THE STMICROELECTRONICS MEETING · CONCEPT

Following that meeting, we conceptualised the [Probe Matrix](#). It is an accessory that places spring probes under the board, so Proto can reach every net to test and program the board without a person moving probes. The concept is at probe-matrix.vercel.app.

3.4 Business documentation

LATE SEPTEMBER · DONE

We prepared a costed bill of materials and cost report. The first [Baker](#) prototype comes to ₹3,73,418 including duty and a 15% contingency, with a projected production cost of ₹1,44,842. We also prepared the [Baker](#) workflow (“the story of a PCB”), a China manufacturing plan, and a pitch deck. Following the round-1 review questions, the plan is now [Baker](#) first, with funding in milestone-based instalments.

What we will do next, one milestone at a time.

The next quarter is about finishing the first [Baker](#) prototype and closing the funding for the proof of concept. [Mixer](#) and the [Probe Matrix](#) stay at design stage until the proof of concept is achieved.

PLANNED STEP	TARGET
Complete the first Baker prototype build	October–November
Run all 15 Baker steps end to end on one board	November–December
Continue software–hardware integration; Proto from a plain-English brief to Gerbers to a Baker job	November–December
Close proof-of-concept funding (STMicroelectronics and others)	October–December

FUNDING PLAN

We are seeking ₹15 lakh for a nine-month proof of concept, or ₹10 lakh in a lean version. We propose it is released in three instalments, each only after the previous milestone is shown: ₹6 lakh at signing; ₹5 lakh once [Baker](#) runs all 15 steps on one board; ₹4 lakh once the process tests pass and Proto produces a [Baker](#) job from a brief. The proof of concept is 20 boards made with Proto + [Baker](#). No founder salary is drawn.

LINKS

WEBSITE	https://protopus-kappa.vercel.app/
BAKER	https://protopus-kappa.vercel.app/#/baker
MIXER	https://protopus-kappa.vercel.app/#/mixer
PROBE MATRIX	https://probe-matrix.vercel.app/

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